

Chapter 5

Experimental Validation of Four-Port SOLR Calibration

Chapter 3 established the experimental foundation for the four-port measurement and validation work that follows by introducing the THz measurement facility, the four-port measurement hardware, the instrument-side calibration hierarchy, and the practical limitations of existing on-chip calibration approaches. Building on that measurement background, Chapter 4 developed the four-port SOLR framework itself, including the error-model basis, the design of the one-port and reciprocal standards, and the definition of the on-chip standard plane after removal of the common probing pads and access network. The present chapter turns from that design-stage and methodology-stage development to measurements on the taped-out calibration die. The chapter aims to determine whether the extracted one-port standards are sufficiently well behaved, repeatable, and distinct to support a stable measured SOLR calibration over the intended frequency range.

A central point of the experimental validation is that the Chapter 4 EM and virtual-mTRL results are retained here as *design references*, but not as absolute ground-truth baselines for the fabricated standards. At mmWave and D-band frequencies, calibrated on-wafer results are known to depend on probe geometry, touchdown conditions, neighboring structures, substrate environment, and reference-plane placement [61, 70, 73]. For that reason, the quality of the fabricated standards is assessed here primarily through measurement-led criteria: physical plausibility of the extracted responses, separation among the standards on the Smith Chart complex reflection plane, repeatability across redundant realizations, conditioning of the resulting SOL solution matrix, and the effect of the extracted standards on the downstream calibration itself [83, 86, 89]. Simulation is used later in the section as a secondary trend check, but not as the sole measure of correctness.

5.1 Measurement and Characterization of the S/O/L Calibration Standards

This section presents the experimental characterization of the fabricated Short, Open, and Load standards after removal of the common on-die embedding between the probe pads and the intended standard-definition plane. Rather than treating the standards as ideal objects whose quality can be judged only by direct comparison to simulation, they are treated here as *measured and extracted calibration standards*. The relevant question is therefore not whether the fabricated responses perfectly reproduce the Chapter 4 design curves, but whether the extraction yields physically plausible, repeatable, and sufficiently distinct one-port definitions at the chosen on-chip reference plane to support a stable SOLR calibration. This interpretation is consistent with prior on-wafer metrology work showing that calibrated results depend not only on nominal standard design, but also on probe geometry, touchdown conditions, neighboring structures, and the exact placement of the reference plane [61, 70, 73].

5.1.1 Measurement Model, Notation, and Extraction Basis

The raw two-port measurements of the fabricated structures are acquired and stored in conventional scattering-parameter form. For any two-port block X , let

$$\mathbf{S}_X(f) = \begin{bmatrix} s_{11,X}(f) & s_{12,X}(f) \\ s_{21,X}(f) & s_{22,X}(f) \end{bmatrix} \quad (5.1)$$

denote its 2×2 scattering matrix, where $s_{ij,X}(f)$ is the scalar scattering parameter from port j to port i , and f is frequency. The corresponding scattering-transfer matrix of the same two-port is denoted by $\mathbf{T}_X(f)$ [100].

Although the measurements are obtained in $\mathbf{S}$ -parameter form, the straight-Thru extraction is written in $\mathbf{T}$ -matrix form because the measured thru is modeled as a physical cascade of sub-networks. In particular, the measured straight Thru is interpreted as a left embedding half-network, an explicit center-line section, and a right embedding half-network. In transfer form, these cascaded sections combine directly through matrix multiplication, which makes extraction and de-embedding of the common embedding substantially more natural than in ordinary $\mathbf{S}$ -parameter form [101–103]. Under the adopted symmetry assumption, the left and right embedding halves are represented by the same transfer matrix $\mathbf{T}_E(f)$, and the measured straight Thru is written as

$$\mathbf{T}_{\text{thru,meas}}(f) = \mathbf{T}_E(f) \mathbf{T}_L(f) \mathbf{T}_E(f), \quad (5.2)$$

where $\mathbf{T}_{\text{thru, meas}}(f)$ is the scattering-transfer matrix of the measured straight Thru, $\mathbf{T}_E(f)$ is the scattering-transfer matrix of one embedding half-network, and $\mathbf{T}_L(f)$ is the scattering-transfer matrix of the explicit center-line section of the straight Thru. In the final extraction baseline, $\mathbf{T}_L(f)$ is constructed from the physical $25.44 \mu\text{m}$ center section, while the remaining pads, vias, and local transitions are absorbed into $\mathbf{T}_E(f)$.

The transmission behavior of the embedding is identified from the measured straight Thru together with the explicit $25.44 \mu\text{m}$ line section. After the initial straight-Thru factorization and bandwise line-model fitting, the de-embedded Thru responses were largely well behaved, but the de-embedded Open and Short remained the main physical concern. However, the measured straight Thru may not, by itself, be sufficiently constraining the reflective behavior at the two sides of the embedding network. In the reflective-refinement stage, the embedding is updated selectively rather than re-estimated in a fully unconstrained way. The transmission terms of the embedding remain anchored to the straight-Thru-derived baseline, while the reflective terms are adjusted using the measured Open, Short, and Load responses. The refinement is guided by physically motivated criteria: the extracted Open and Short should remain strongly reflective and near the unit circle, the extracted Load should remain comparatively small, the updated embedding should remain smooth across frequency, and the final result should stay close to the original thru-derived baseline unless the measured SOL data provides clear evidence to move it.

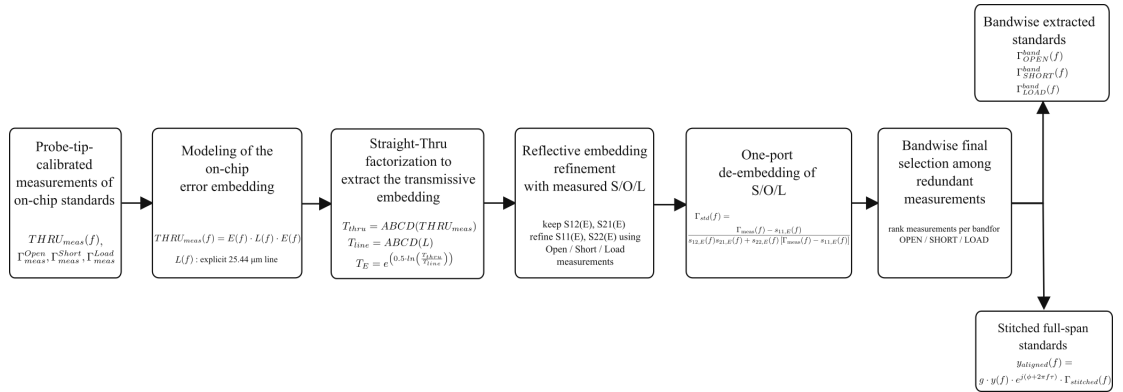


Figure 5.1: One-port standard extraction procedure: probe-tip-calibrated measurement of the fabricated straight Thru and one-port standards, identification of the common embedding from the straight Thru with the explicit $25.44 \mu\text{m}$ center-line model, reflective refinement using measured Open/Short/Load data, de-embedding to the common standard-definition plane, and generation of both bandwise extracted and stitched full-span standard representations.

The next stage of the extraction therefore treated the one-port standards not merely

as outputs of the de-embedding, but also as additional measurements used to improve the reflective part of the embedding model. The reflective behavior is then refined using the measured Open, Short, and Load responses, since the straight Thru alone constrains the transmission terms more strongly than the reflection terms. The final de-embedding procedure anchors the forward and reverse transmission behavior to the thru-derived solution, while the reflective terms are adjusted so that the extracted one-port standards become more physically consistent at the intended standard plane. The transmission-led embedding extraction followed by reflective refinement is well aligned with prior calibration-kit and de-embedding practice for mmWave integrated-circuit measurements [101, 103].

For a one-port standard, let $\Gamma_{\text{meas}}(f)$ denote the measured reflection coefficient at the probe-tip-calibrated plane, and let $\Gamma_{\text{std}}(f)$ denote the extracted reflection coefficient after de-embedding to the common standard-definition plane. Using the scattering description of the embedding half-network,

$$\mathbf{S}_E(f) = \begin{bmatrix} s_{11,E}(f) & s_{12,E}(f) \\ s_{21,E}(f) & s_{22,E}(f) \end{bmatrix}, \quad (5.3)$$

the extracted one-port response is written as

$$\Gamma_{\text{std}}(f) = \frac{\Gamma_{\text{meas}}(f) - s_{11,E}(f)}{s_{12,E}(f)s_{21,E}(f) + s_{22,E}(f)[\Gamma_{\text{meas}}(f) - s_{11,E}(f)]}. \quad (5.4)$$

Here, $s_{11,E}(f)$, $s_{12,E}(f)$, $s_{21,E}(f)$, and $s_{22,E}(f)$ are the scalar elements of the embedding half-network scattering matrix $\mathbf{S}_E(f)$. Equation (5.4) removes the direct front-face reflection of the embedding and the forward/reverse loading through the embedding network, yielding frequency-dependent definitions of $\Gamma_{\text{open}}(f)$, $\Gamma_{\text{short}}(f)$, and $\Gamma_{\text{load}}(f)$ at the same on-chip plane later used by the SOLR solver.

Two forms of the extracted standards are retained in the discussion that follows. The first is the *accepted bandwise extraction*, which is the primary characterization result within each measurement band after comparison of redundant realizations. The second is a *stitched full-span representation*, which provides a smooth cross-band view for reporting and downstream calibration use. The stitched curve is therefore interpreted as a regularized aggregate of the accepted bandwise measurements rather than as a replacement for them.

A final point of interpretation is that no hard passivity projection is imposed on the accepted extracted one-port standards. If mild non-passive behavior appears after de-embedding, it is treated as a diagnostic of residual model limitation or measurement uncertainty rather than being hidden by a final projection step. This choice preserves

the distinction between *characterizing the fabricated standard* and *post-correcting it into an idealized response*.

5.1.2 Evaluation Framework for the Extracted Standards

Because the standards are obtained through a measurement-led extraction rather than by direct ideal assignment, their quality is assessed here through the following complementary criteria:

1. **Physical plausibility.** The extracted standards should remain physically interpretable at the adopted reference plane. The Open and Short are expected to remain strongly reflective and to follow smooth capacitive and inductive trajectories, respectively, on the Smith chart. The Load should remain near the chart center over the usable frequency range, with positive real impedance and limited reactance [61, 70].
2. **Separation and identifiability.** The Open, Short, and Load must remain sufficiently distinct in the complex reflection plane so that the one-port SOL system remains well posed. If the extracted Load drifts too close to either reflective standard, or if the Open and Short collapse toward similar trajectories, the solved error terms become increasingly sensitive to measurement noise and de-embedding error [103, 104].
3. **Repeatability across redundant realizations.** Since the one-port standards were measured redundantly from opposite sides of the embedding, the spread among those extractions becomes part of the calibration-quality assessment rather than a secondary detail. Small spread indicates that the extracted standard is primarily determined by the physical structure, while large spread indicates residual sensitivity to embedding uncertainty, touchdown variation, or model incompleteness [61, 73].
4. **Extraction consistency.** The extracted standards should remain consistent with the underlying thru-derived embedding model. In practice, this means that reflective refinement should improve the physical plausibility of the Open, Short, and Load without materially degrading the transmission behavior already established by the straight Thru [101, 103].
5. **Numerical conditioning of the one-port SOL solve.** The extracted standards are ultimately used as the known reflection coefficients in the one-port three-term calibration model. Let $\Gamma_k(f)$ denote the extracted reflection coefficient of

standard $k \in \{O, S, L\}$ at the standard-definition plane, and let $\Gamma'_k(f)$ denote the corresponding measured reflection coefficient at the probe-tip-calibrated plane. Using the standard three-term one-port model, the unknown error terms $e_{00}(f)$, $e_{11}(f)$, and $\Delta_e(f)$ satisfy

$$\mathbf{A}_{\text{SOL}}(f) \mathbf{x}_{\text{SOL}}(f) = \mathbf{b}_{\text{SOL}}(f), \quad (5.5)$$

where

$$\mathbf{x}_{\text{SOL}}(f) = \begin{bmatrix} e_{00}(f) \\ e_{11}(f) \\ \Delta_e(f) \end{bmatrix}, \quad \mathbf{b}_{\text{SOL}}(f) = \begin{bmatrix} \Gamma'_O(f) \\ \Gamma'_S(f) \\ \Gamma'_L(f) \end{bmatrix}, \quad (5.6)$$

and

$$\mathbf{A}_{\text{SOL}}(f) = \begin{bmatrix} 1 & \Gamma_O(f)\Gamma'_O(f) & -\Gamma_O(f) \\ 1 & \Gamma_S(f)\Gamma'_S(f) & -\Gamma_S(f) \\ 1 & \Gamma_L(f)\Gamma'_L(f) & -\Gamma_L(f) \end{bmatrix}. \quad (5.7)$$

Here, $\mathbf{A}_{\text{SOL}}(f)$ is the 3×3 coefficient matrix of the one-port SOL solve, $\mathbf{x}_{\text{SOL}}(f)$ is the vector of one-port error terms, and $\mathbf{b}_{\text{SOL}}(f)$ is the measured-data vector. The robustness of this solve is evaluated through the 2-norm condition number

$$\kappa_{\text{SOL}}(f) = \kappa_2(\mathbf{A}_{\text{SOL}}(f)) = \|\mathbf{A}_{\text{SOL}}(f)\|_2 \|\mathbf{A}_{\text{SOL}}^{-1}(f)\|_2 = \frac{\sigma_{\max}(\mathbf{A}_{\text{SOL}}(f))}{\sigma_{\min}(\mathbf{A}_{\text{SOL}}(f))}, \quad (5.8)$$

where $\sigma_{\max}(\cdot)$ and $\sigma_{\min}(\cdot)$ are the largest and smallest singular values, respectively. A small value of $\kappa_{\text{SOL}}(f)$ indicates that the solved error terms are relatively insensitive to small perturbations in the extracted standards, while a large value indicates that small errors in the Open, Short, or Load can be strongly amplified in the resulting calibration [104, 105].

6. **Model closure residual.** After solving for $e_{00}(f)$, $e_{11}(f)$, and $\Delta_e(f)$, the corresponding reconstructed measured reflection coefficient of standard k is

$$\widehat{\Gamma}'_k(f) = \frac{e_{00}(f) - \Delta_e(f)\Gamma_k(f)}{1 - e_{11}(f)\Gamma_k(f)}, \quad k \in \{O, S, L\}, \quad (5.9)$$

and the closure residual is defined as

$$r_k(f) = \left| \Gamma'_k(f) - \widehat{\Gamma}'_k(f) \right|. \quad (5.10)$$

Here, $\widehat{\Gamma}'_k(f)$ is the reconstructed measured reflection coefficient predicted by the solved one-port model, and $r_k(f)$ is the corresponding scalar residual magnitude.

This residual provides an internal lack-of-fit check for the one-port calibration model. In the present work, it is interpreted together with $\kappa_{\text{SOL}}(f)$: a small residual is most meaningful when the underlying system is also well conditioned [104, 105].

7. **Simulation comparison as a secondary reference.** Comparison to the Chapter 4 EM and virtual-mTRL results is retained as a trend-level reference only. It remains useful to verify that the measured Open, Short, and Load follow the broad behavior anticipated in Chapter 4, such as the direction of Smith-chart rotation and the approximate onset of Load degradation. However, because both the extracted reference plane and the fabricated probing environment differ from the idealized EM setup, disagreement with simulation is interpreted cautiously and is not by itself treated as evidence of extraction failure [61, 70, 73].

Table 5.1: Proposed analyses for evaluating the extracted one-port standards.

Metric	Purpose	Result Placeholder
Physical plausibility	Verify smooth and interpretable Open/Short/Load behavior at the extracted reference plane	[TBD]
Standard separation	Confirm that Open, Short, and Load remain sufficiently distinct in the complex Γ -plane	[TBD]
Redundant-measurement repeatability	Quantify spread across redundant realizations and repeated touchdowns	[TBD]
Extraction consistency	Assess whether reflective refinement improves one-port behavior while remaining consistent with the thru-derived embedding	[TBD]
SOL solve conditioning	Evaluate $\kappa_{\text{SOL}}(f) = \kappa_2(\mathbf{A}_{\text{SOL}}(f))$ versus frequency	[TBD]
Closure residual	Evaluate $r_k(f) = \Gamma'_k(f) - \widehat{\Gamma}'_k(f) $ for $k \in \{\text{O}, \text{S}, \text{L}\}$	[TBD]
Simulation trend agreement	Check broad trend agreement without using simulation as absolute truth	[TBD]

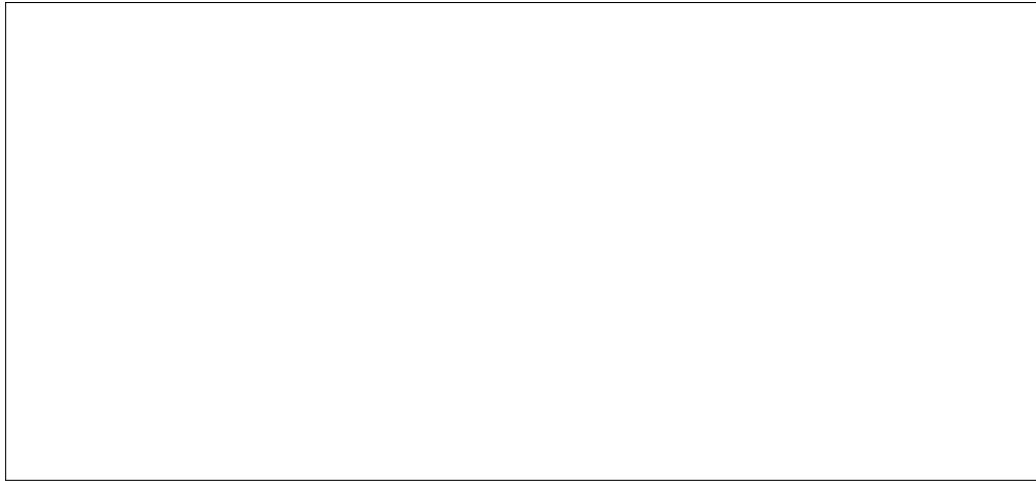


Figure 5.2: Placeholder for a figure showing, for each of the Open, Short, and Load standards, the accepted bandwise extracted data together with the stitched full-span representation used for continuity and downstream calibration.

5.1.3 Measured Open and Short Standards

The Open and Short serve as the strongly reflective standards of the one-port set and are therefore judged primarily by their smoothness, repeatability, and distinctness at the extracted reference plane. The Open is expected to trace a predominantly capacitive trajectory on the Smith chart, while the Short is expected to trace a predominantly inductive trajectory, with both remaining close to the unit circle over most of the usable band. The central experimental question is not whether either standard perfectly overlays the Chapter 4 design reference, but whether the extracted responses remain physically plausible and sufficiently distinct to support a robust reflectometer solve [61, 103].

The following analyses should be reported for the Open and Short:

- **Smith-chart trajectory and reflection magnitude.** Plot the accepted bandwise and stitched representations of the extracted Open and Short on the Smith chart and report $|\Gamma_{\text{open}}|$ and $|\Gamma_{\text{short}}|$ versus frequency.

Result placeholder: The extracted Open remained above _____ in $|\Gamma|$ up to _____ GHz, while the extracted Short remained above _____ up to _____ GHz. Both standards exhibited _____ Smith-chart trajectories, with the first noticeable irregularities appearing near _____ GHz.

- **Effect of reflective refinement.** Compare the final extracted Open and Short

against the earlier thru-anchored extraction to show whether reflective refinement improves physical plausibility.

Result placeholder: Relative to the earlier thru-only extraction, the final extraction produced _____ improvement in the Open and Short responses, most clearly visible in _____.

- **Redundant-measurement agreement.** Overlay redundant realizations where available and document the selected representation in each band.

Result placeholder: For the Open, the agreement between redundant realizations remained within _____ in magnitude and _____ in phase below _____ GHz. For the Short, the corresponding spread was _____.

- **Contribution to one-port conditioning.** Relate the continued separation of the Open and Short from each other and from the Load to the behavior of $\kappa_{\text{SOL}}(f)$.

Result placeholder: Across the frequency range _____, the Open and Short remained sufficiently separated from each other and from the Load such that $\kappa_{\text{SOL}}(f)$ stayed below _____. Above _____ GHz, the conditioning degraded primarily because _____.

- **Comparison to Chapter 4 trends.** Include measured-versus-simulated overlays as secondary references only.

Result placeholder: Relative to the Chapter 4 design reference, the measured Open and Short showed _____ agreement in trajectory and magnitude trend, while the main discrepancy was _____.

5.1.4 Measured Load Standard

The Load is the most critical one-port standard in the kit because it determines how well the one-port system remains anchored near the center of the Smith chart. Unlike the Open and Short, the Load is not judged only by visual proximity to $\Gamma = 0$. A practically useful extracted Load may remain imperfect, provided that it stays sufficiently close to the center of the Smith chart, sufficiently repeatable across redundant realizations,

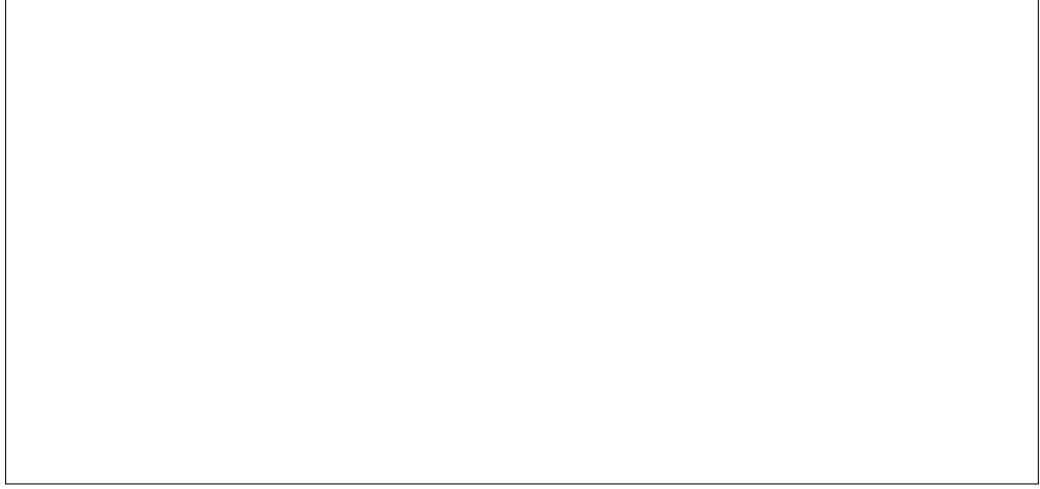


Figure 5.3: Placeholder for a figure showing the extracted Open and Short responses, for example: (a) Smith-chart loci of accepted bandwise and stitched results, (b) $|\Gamma|$ versus frequency, (c) phase versus frequency, and (d) comparison between earlier thru-only extraction, final extraction baseline, and Chapter 4 design-reference simulation.

and sufficiently distinct from the reflective standards to keep the one-port solve well conditioned [103,104]. In that sense, the useful bandwidth of the one-port kit is expected to be set primarily by the extracted Load.

The following analyses should be reported for the Load:

- **Reflection magnitude and extracted impedance.** Report $|\Gamma_{\text{load}}(f)|$ together with the extracted impedance

$$Z_{\text{load}}(f) = Z_0 \frac{1 + \Gamma_{\text{load}}(f)}{1 - \Gamma_{\text{load}}(f)}, \quad (5.11)$$

where Z_0 is the reference impedance of the measurement system.

Result placeholder: The extracted Load remained below _____ in $|\Gamma|$ up to _____ GHz. Over this range, $\Re\{Z_{\text{load}}(f)\}$ remained between _____ and _____ Ω , while $\Im\{Z_{\text{load}}(f)\}$ stayed within _____ Ω .

- **Redundant-measurement repeatability.** Quantify the spread of the extracted Load across redundant realizations.

Result placeholder: The redundant-measurement spread of the extracted Load remained within _____ below _____ GHz, increasing to _____ near the upper edge of the band.

- **Impact on one-port conditioning.** Evaluate $\kappa_{\text{SOL}}(f)$ together with the Load observables to identify the frequency region where the Load ceases to remain sufficiently distinct from the reflective standards.

Result placeholder: The one-port system remained well conditioned up to _____ GHz, with $\kappa_{\text{SOL}}(f)$ below _____. A rapid increase occurred near _____ GHz and coincided with _____ in the extracted Load response.

- **Model closure residual.** Evaluate $r_{\text{L}}(f)$ using (5.10).

Result placeholder: The mean closure residual for the Load was _____ below _____ GHz and increased to _____ above _____ GHz, indicating that _____.

- **Comparison to Chapter 4 trends.** Compare the measured Load to the Chapter 4 design reference only at the trend level.

Result placeholder: The measured Load qualitatively followed the Chapter 4 design trend by remaining closest to the Smith-chart center below approximately _____ GHz and then drifting toward _____ behavior at higher frequency.

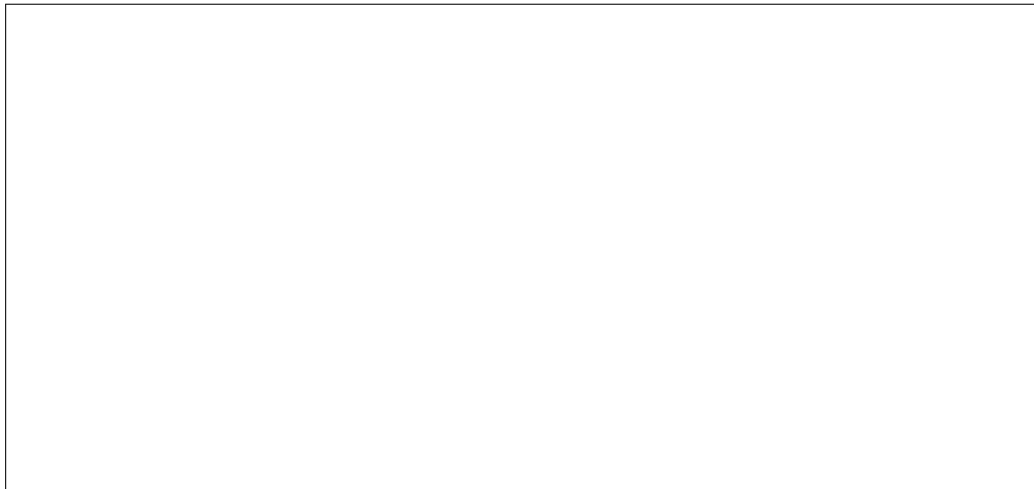


Figure 5.4: Placeholder for a figure showing the extracted Load response, for example: (a) Smith-chart locus of accepted bandwise and stitched results, (b) $|\Gamma_{\text{load}}|$ versus frequency, (c) extracted R and X of Z_{load} , and (d) $\kappa_{\text{SOL}}(f)$ and closure-residual metrics plotted against frequency.

5.1.5 Adopted One-Port Standards for the Measured SOLR Calibration

Based on the analyses above, the relevant outcome for the remainder of the chapter is not whether the fabricated standards are ideal, but whether the extracted one-port definitions are sufficiently trustworthy for the measured SOLR calibration to proceed with confidence. The Open and Short must remain strongly reflective, smooth, and distinct, while the Load must remain sufficiently central and repeatable to keep the one-port system well posed. In this sense, the practical upper frequency limit of the one-port kit is determined by the measured Load behavior together with the growth of $\kappa_{\text{SOL}}(f)$ and the closure residual, rather than by simulation alone [61, 103, 104].

For the downstream SOLR implementation, the accepted bandwise standards are retained as the primary characterized one-port definitions. The stitched full-span curves are retained as complementary reporting products and as convenient continuous reference views across bands. They are not interpreted as exact replacements for the accepted bandwise measurements. Likewise, any residual mild non-passivity in the accepted extracted standards is retained as part of the measured characterization and carried forward transparently.

Result placeholder: From the combined plausibility, repeatability, conditioning, and closure analyses, the extracted one-port standards are judged reliable for subsequent SOLR use up to _____ GHz. Within this range, the dominant limitation is _____. Beyond this range, the calibration is expected to become increasingly sensitive to _____.

Table 5.2: Summary of the adopted one-port standards for the measured SOLR calibration.

Standard	Primary Representation	Selected Re-	Usable Range	Notes
Open	[Bandwise Stitched]	/ [TBD]	[TBD]	[TBD]
Short	[Bandwise Stitched]	/ [TBD]	[TBD]	[TBD]
Load	[Bandwise Stitched]	/ [TBD]	[TBD]	[TBD]